

▪ Declaring and Accessing Arrays

 **Arrays:** create a group of variables *with the same data type*.

- In an array
 - Each element behaves like a variable.
 - All elements must have the same data type.
 - Elements either can contain primitive data or can be reference variables.
- Arrays can be either one - dimensional or multidimensional.
- **One - dimensional** array consists of elements arranged in a row.
- **Two - dimensional** array has *both* rows and columns.
- **Three - dimensional** array has rows, columns, and pages.
- VB .NET implements multi-dimensional arrays as arrays of arrays.

➤ Using One-Dimensional Arrays

✓ Declare a 5-element array with of integers:

' declare an integer array with 5 elements

```
Dim A(5) As Integer
```

✓ Individual array elements are accessed by writing the array reference variable, followed by the index value of the element enclosed in parentheses.

✓ Code to initialize the array elements:

```
A(1) = 75
```

```
A(2) = 80
```

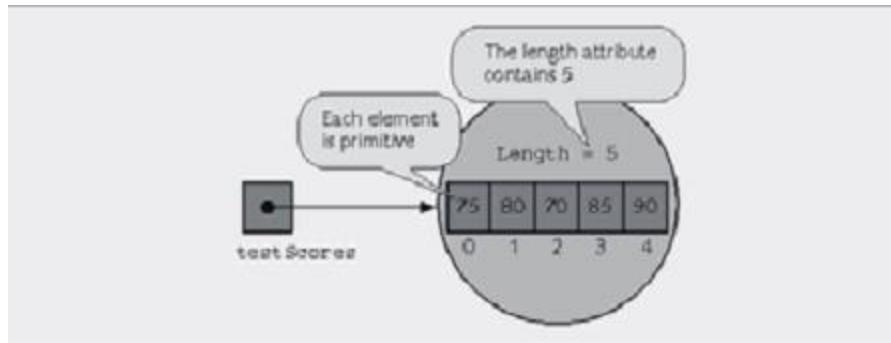
```
A(3) = 70
```

```
A(4) = 85
```

```
A(5) = 90
```

- ✓ An array can be declared and populated using a single statement:

Dim A() As Integer = {75, 80, 70, 85, 90}



A five elements Integer Array

Example 1:

Write a program in VB. Net to read array X(10) then find the summation of all elements in this array using Do ... While statement?

Writing the Code

Use the Code Editor

- Double-click the Button1 on the form and write the following code inside it:

```
Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
```

```
    Dim I As Integer = 1
```

```
    Dim X(10) As Integer
```

```
    Dim Y As Integer = 0
```

```
    Do While I <= 10
```

```
        X(I) = InputBox("Enter The Value of Array X ")
```

```
        Y = Y + X(I)
```

```
        I = I + 1
```

```
    Loop
```

```
    MessageBox.Show("The Summation is: " & Y)
```

```
End Sub
```

Example 2:

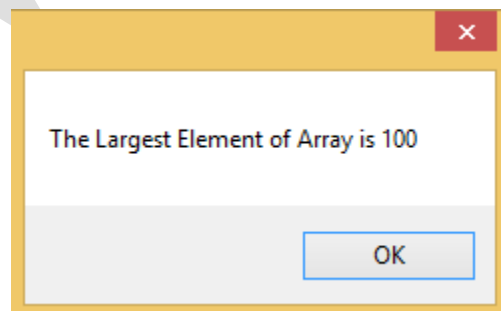
Write a program in VB. Net to read array B(5) then find the largest element in this array using For ... Next statement?

Writing the Code

Use the Code Editor

- Double-click the Button1 on the form and write the following code inside it:

```
Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
    Dim I As Integer = 0
    Dim B() As Integer = {5, 80, 100, 60, 30}
    Dim Max As Integer = B(0)
    For I = 1 To 4
        If B(I) > Max Then
            Max = B(I)
        End If
    Next
    MessageBox.Show("The Largest Element of Array is " & Max)
End Sub
```



Example 3:

Write a program in VB. Net to read array M(10) then multiply each element in this array by (5) using For .. Next Statement?

Writing the Code

Use the Code Editor

- Double-click the Button1 on the form and write the following code inside it:

```
Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
```

```
    Dim I As Integer
```

```
    Dim M(10) As Integer
```

```
    Dim W(10) As Integer
```

```
    For I = 1 To 10
```

```
        M(I) = InputBox("Enter The Values of Array M ")
```

```
        W(I) = M(I) * 5
```

```
    Next
```

```
    For I = 1 To 10
```

```
        Console.Write(M(I) & " ")
```

```
    Next
```

```
    Console.WriteLine()
```

```
    Console.WriteLine()
```

```
    For I = 1 To 10
```

```
        Console.Write(W(I) & " ")
```

```
    Next
```

```
End Sub
```

Example4:

Design the following form and write a program in VB. Net to read two arrays A(5), B(5) then find the summation of elements between arrays A and B in new array C(5) using For .. Next Statement?

The Solution is:

To design the above form we implement the following set properties of the objects:

Object	Property	Setting
Button1	Text	Sum
Button2	Text	Clear
Button3	Text	Exit
Label1	Text	Array A
Label2	Text	Array B
Label3	Text	Array C
Form1	Text	Summation of Two Arrays
Form1	MaximizeBox	False
Form1	MinimizeBox	False
ListBox1	Items.add()	Elements of A(5)

ListBox2	Items.add()	Elements of B(5)
ListBox3	Items.add()	Elements of C(5)

Writing the Code

Use the Code Editor

1. Double-click the Sum Button1 on the form and write the following code inside it:

```
Private Sub Button1_Click_1(sender As Object, e As EventArgs) Handles Button1.Click
    Dim I As Integer
    Dim A(5) As Integer
    Dim B(5) As Integer
    Dim C(5) As Integer
    For I = 1 To 5
        A(I) = InputBox("Enter The Elements of Array A")
        ListBox1.Items.Add(A(I))
    Next
    For I = 1 To 5
        B(I) = InputBox("Enter The Elements of Array B")
        ListBox2.Items.Add(B(I))
    Next
    For I = 1 To 5
        C(I) = A(I) + B(I)
        ListBox3.Items.Add(C(I))
    Next
End Sub
```

2. Double-click the Clear Button2 on the form and write the following code inside it:

```
Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
    ListBox1.Items.Clear()
    ListBox2.Items.Clear()
    ListBox3.Items.Clear()
End Sub
```

3. Double-click the Exit Button3 on the form and write the following code inside it:

```
Private Sub Button3_Click(sender As Object, e As EventArgs) Handles Button3.Click
    Me.Close()
End Sub
```

▪ Using Multidimensional Arrays

➤ Conceptually

- A two - dimensional array is like a table with rows and columns.
- A three - dimensional array is like a cube, with rows, columns, and pages.
- Each dimension has its own index.
- Declare an Integer array with five rows and two columns.

Example:

```
Dim A(3, 3) As Integer
```

Example 1:

Write a program in VB. Net to read array A(3, 3) then print this array using For .. Next Statement?

Writing the Code

Use the Code Editor

- Double-click the Button1 on the form and write the following code inside it:

```
Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
```

```
    Dim I, J As Integer
```

```
    Dim A(3, 3) As Integer
```

```
    For I = 1 To 3
```

```
        For J = 1 To 3
```

```
            A(I, J) = InputBox("Enter The Values of Array A ")
```

```
        Next
```

```
    Next
```

```
    For I = 1 To 3
```

```
        For J = 1 To 3
```

```
            Console.Write(A(I, J) & " ")
```

```
        Next
```



```
Console.WriteLine()  
Next  
End Sub
```

Example 2:

Write a program in VB. Net to read array B(3, 3) then find the largest element in this array using For ... Next statement?

Writing the Code

Use the Code Editor

- Double-click the Button1 on the form and write the following code inside it:

```
Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click  
    Dim I, J As Integer  
    Dim B(3, 3) As Integer  
    Dim Max As Integer = B(1, 1)  
    For I = 1 To 3  
        For J = 1 To 3  
            B(I, J) = InputBox("Enter The Values of Array A ")  
        Next  
    Next  
    For I = 1 To 3  
        For J = 1 To 3  
            If B(I, J) > Max Then  
                Max = B(I, J)  
            End If  
        Next  
    Next  
    MessageBox.Show("The Largest Element is " & Max)  
End Sub
```

Example 3:

Write a program in VB. Net to read array B(3, 3) then multiply each element in this array by (5) using For .. Next Statement?

Writing the Code

Use the Code Editor

- Double-click the Button1 on the form and write the following code inside it:

```
Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
    Dim I, J As Integer
    Dim B(3, 3) As Integer
    Dim C(3, 3) As Integer
    For I = 1 To 3
        For J = 1 To 3
            B(I, J) = InputBox("Enter The Values of Array A ")
            C(I, J) = B(I, J) * 5
        Next
    Next
    For I = 1 To 3
        For J = 1 To 3
            Console.Write(B(I, J) & " ")
        Next
        Console.WriteLine()
    Next
    Console.WriteLine()
    For I = 1 To 3
        For J = 1 To 3
            Console.Write(C(I, J) & " ")
```

Next

Console.WriteLine()

Next

End Sub

Homework 1:

Write a program in VB. Net to read two arrays A(3, 3), B(3,3) then find the summation of elements between arrays A and B in new array C(3,3) using For .. Next Statement?

Homework 2:

Write a program in VB. Net to read two arrays A(3, 3), B(3,3) then find the Multiplication of elements between arrays A and B in new array C(3,3) using For .. Next Statement?